Sandbox Scenario Comprehensive Case

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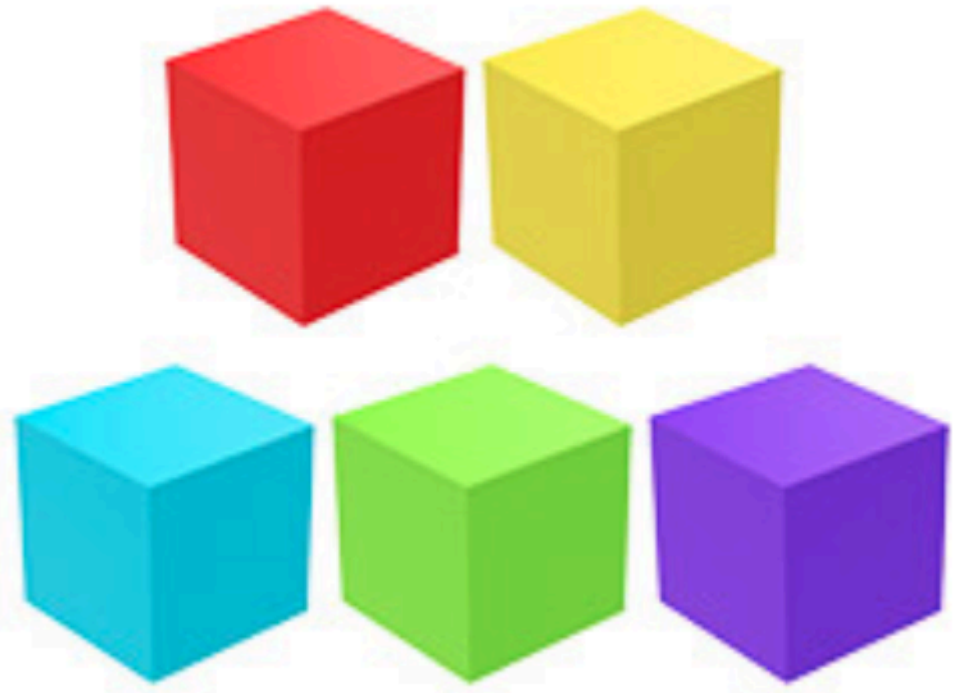
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1. Course Content

- Through the study of the preceding lessons, we have used the slam_toolbox algorithm to build a high-precision map and pose graph for the sandbox world, deployed a "visual relocalization lighthouse" that never drifts, and marked a clear semantic place name for each workstation.
- The robot needs to have the ability to execute **ultra-long-period composite tasks** lasting tens of minutes:
- It no longer moves a finger upon hearing a single command; instead, it receives a macroscopic enterprise production scheduling plan. It needs to autonomously travel across the large sandbox map, autonomously recognize, sort, quality-inspect, and transport across areas between multiple workshop workstations, and automatically return to the parking spot to stand by after all tasks are completed.
- This fully demonstrates OpenClaw's powerful task orchestration and closed-loop execution capability as the **"workshop autonomous operation brain"**!

2. Preparation

This case uses the included 4 * 4 * 4 cm foam color blocks and 3 * 3 * 3 cm waste pattern blocks.

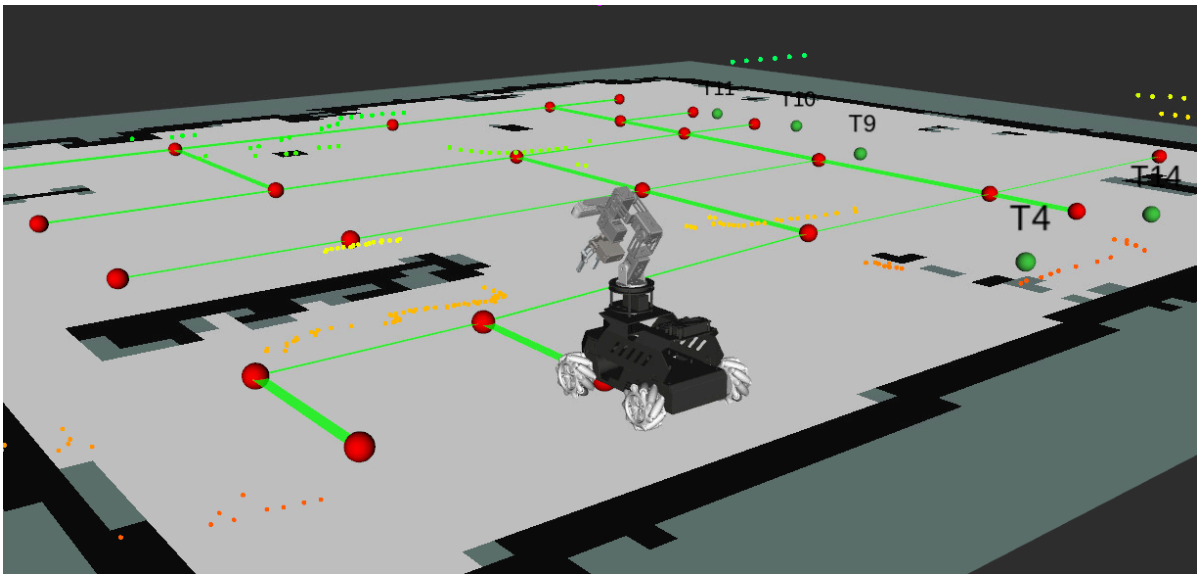




2.1 Complete the Preceding Lessons

- Complete the [Building the Sandbox Map], [Visual Relocalization Marker], and [Map Mapping Marker] lessons.

```
ros2 launch yahboomcar_bringup mcp_service.launch.py use_camera:=true
use_nav:=true relocalization:=true use_waste_classify:=true
```

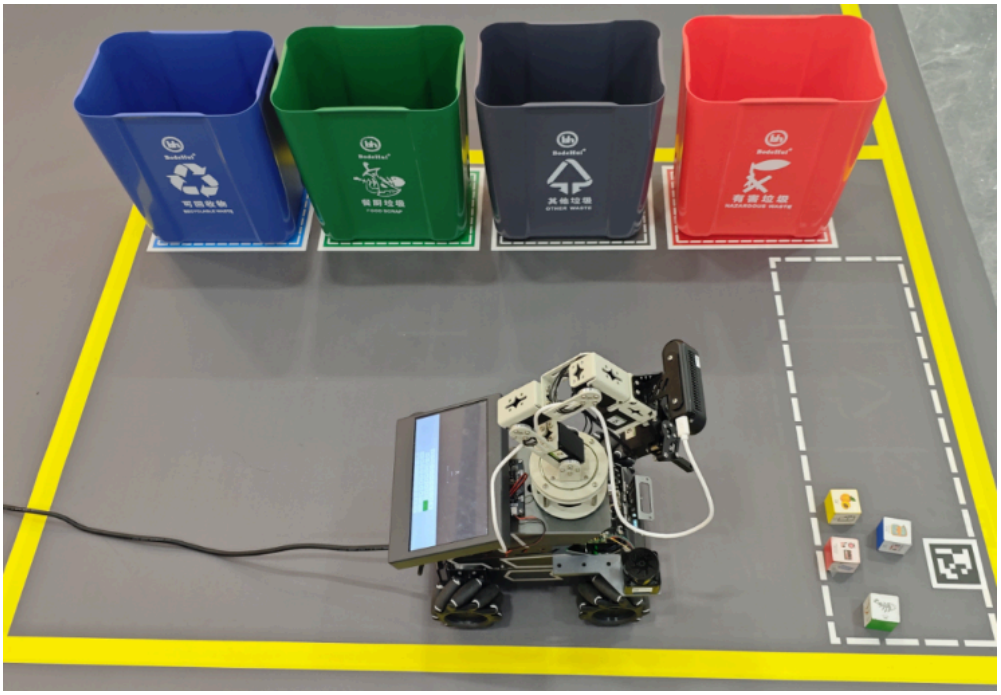


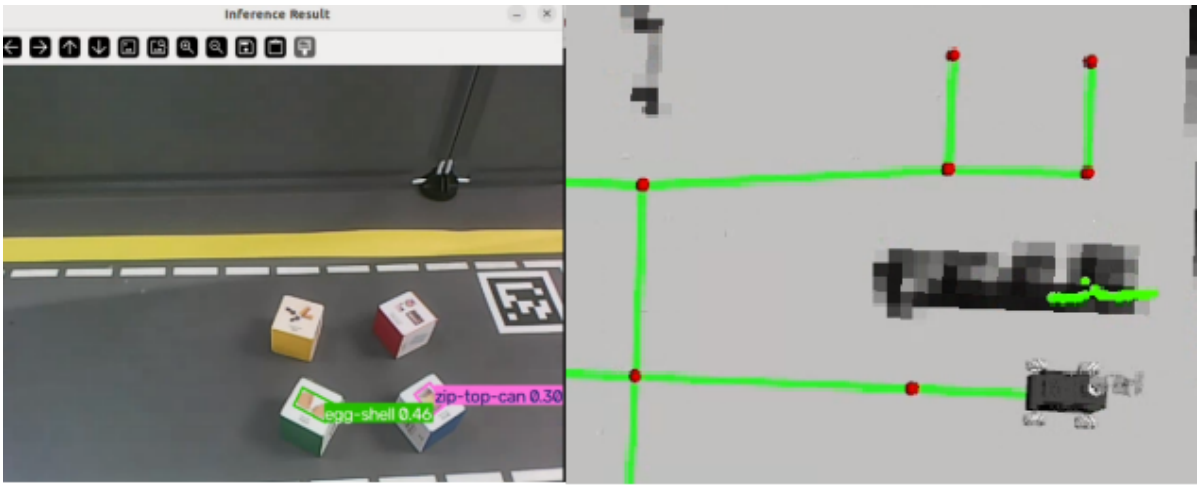
3. Reference Cases

> **Tip**
>
> - You can customize your own instruction cases and gameplay functions according to actual needs. The following are some reference cases.

3.1 Waste Sorting

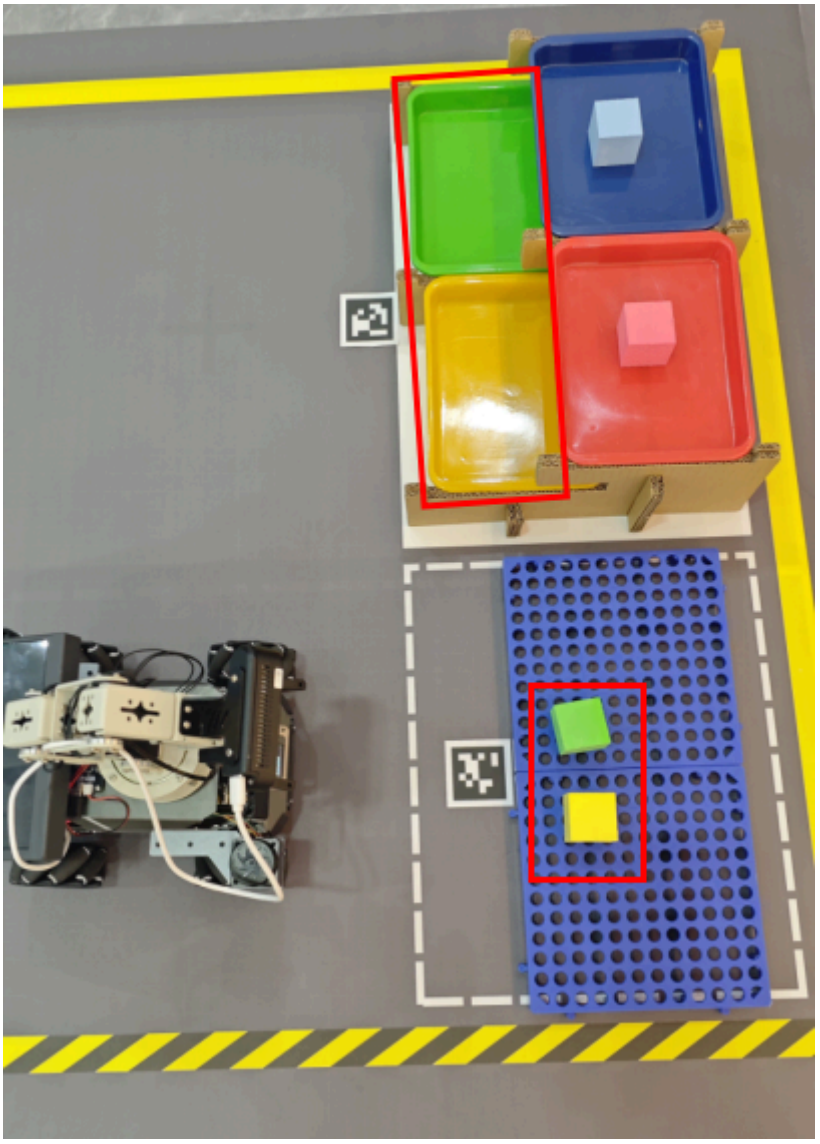
- Go to the waste sorting area and put the recyclable waste and kitchen waste into the corresponding trash cans.





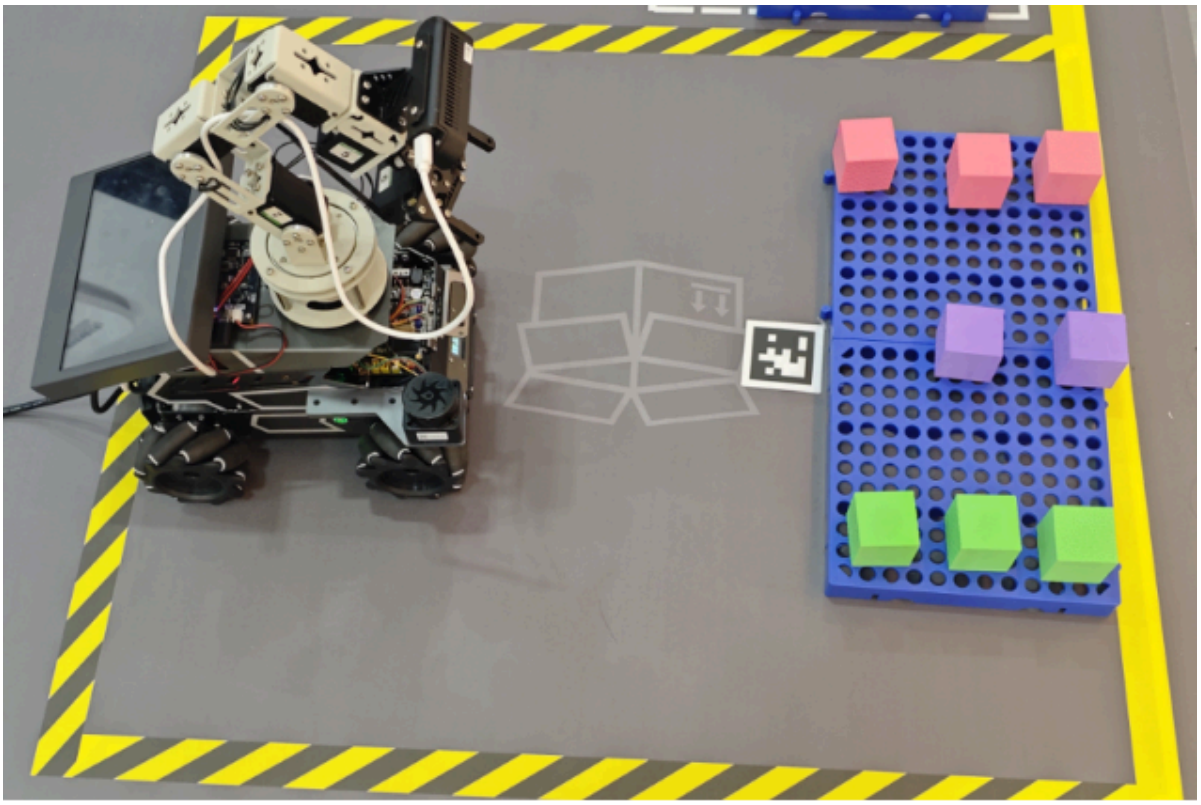
3.2 Goods Sorting

- Go to the goods sorting area and sort all the goods of every color on the tray onto the shelves of the corresponding colors.

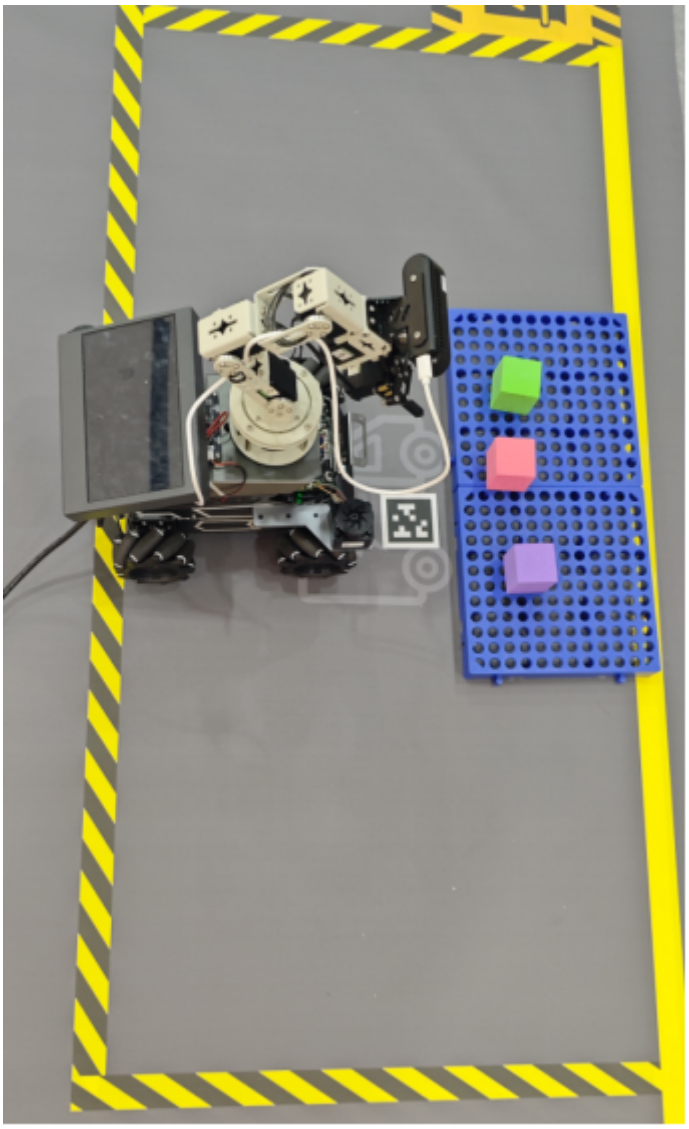


3.3 Goods Inspection and Replenishment

- Check the goods missing in the shipping area and replenish them from the goods warehouse.
- Goods warehouse

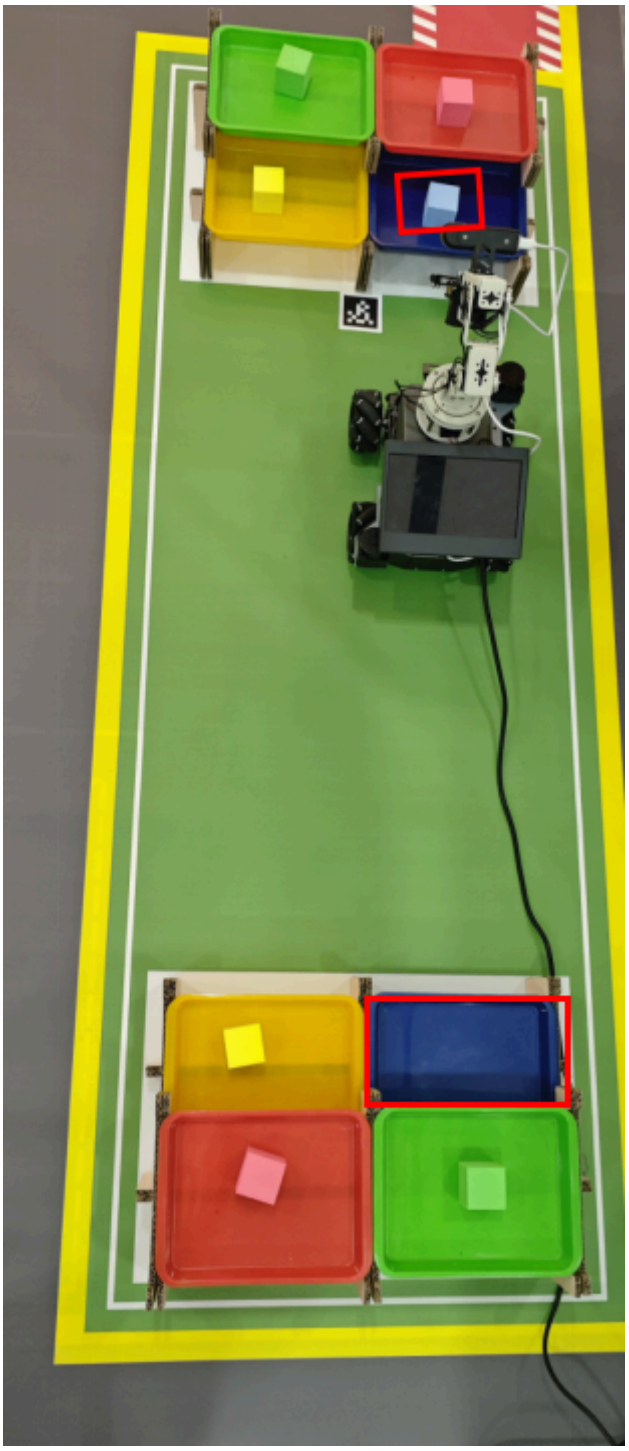


- Shipping area



3.4 Transporting Objects Between Shelves

- Transport the blue goods from Shelf No. 1 to the corresponding tray of Shelf No. 2.



4. Comprehensive Case

- **You can combine the above cases in any way**, or customize your own instruction cases, writing instructions according to actual needs. The following is a reference comprehensive case:
- Factory comprehensive case 1

> **Important**
>
> - The long-process task shown here takes a relatively long time, about 25-40 minutes.

```
Start today's factory inspection. First sort all the waste in the waste area into the corresponding trash cans, then go to the goods sorting area and sort the goods onto the areas of the corresponding colors on Shelf No. 3. After that, go to the shipping area to see whether any goods are missing; if goods are missing, replenish them from the goods warehouse. Finally, take a blue item from Shelf No. 1 and deliver it to the tray of the corresponding color on Shelf No. 2. Return to parking lot No. 1 after the task is complete.
```

- Just choose any one of the interaction methods for sending the instruction.